

Tutorial and Practical work n° 2

Level: 1st Year of the Foundation training

Material: Algorithms and Static Data Structures

Exercise 01: Positive/Negative product

Write an algorithm that checks if the product of two numbers and displays whether the product is positive, negative, or zero.

Example:

M=4, N=-3, Product=-12 is negative.

M=0, N=15, Product=0 is zero.

Exercise 02: Calculate Trip Duration

Write an algorithm that calculates the total duration of a trip, given the departure and arrival times in hours and minutes.

Example:

Departure time = 9h00, Arrival time = 11h30: Total trip duration = 2h30

Departure time = 22h30, Arrival time = 1h15: Total trip duration = 2h45

Exercise 03: VAT calculation

Given an amount excluding VAT and a VAT rate of 20%, write an algorithm that calculates the total amount including VAT. In case the amount including VAT exceeds 2000 DA, apply a discount on the total of 15%.

Example: Amount excluding VAT = 1800 DA, VAT = 360 DA, 2160 > 2000, apply a 15% discount = 324 DA, Final price = 2160 - 324 = 1836 DA.

Exercise 04: Simple calculator

Write an algorithm that simulates a simple calculator, given two numbers: +, -, *, /. Handle the case of division by zero.

Example: 10 + 5 = 15.

9 * 3 = 27.

8 / 0 = Error.

Exercise 05: Number of the Month

Write an algorithm that displays the month corresponding to a number between 1 and 12.

Example:

number = 5, month = May

number = 13, Invalid number

Exercise 07: Count the number of digits

Write an algorithm that counts the number of digits an integer n contains.

Exercise 08: Sum of digits of n

Write an algorithm that computes the sum of all digits of a number n .

Example: $n = 4321 \rightarrow 4 + 3 + 2 + 1 = 10$.

Exercise 09: Sum of squares

Given a positive integer n , write an algorithm to find the sum of squares of first n integer numbers.

Example

Input: $n = 2$, **Output:** 5, Explanation: $1^2 + 2^2 = 5$

Input: $n = 8$, **Output:** 204, Explanation: $1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 = 204$

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Exercise 10: Swap two numbers

Given two numbers a and b , write an algorithm to swap them.

Example

Input: $a = 2, b = 3$, **Output:** $a = 3, b = 2$

Input: $a = 20, b = 0$, **Output:** $a = 0, b = 20$

Input: $a = 10, b = 10$, **Output:** $a = 10, b = 10$

Exercise 11: Distance between two points

Write an algorithm to calculate the distance between two points $p_1(x_1, y_1)$ and $p_2(x_2, y_2)$.

Example

Input: $p_1(x_1, y_1) = (3, 4), p_2(x_2, y_2)$, **Output :** 5

Input: $p_1(x_1, y_1) = (3, 4), p_2(x_2, y_2)$, **Output :** 1.41421

Exercise 12: Reverse a number

Write an algorithm that reverses the digits of a number n .

Example: $n = 1234 \rightarrow 4321$.

Exercise 13: Sum of even and odd digits

Write an algorithm that calculates separately for an integer number n , the sum of even digits, and the sum of odd digits.

Example: $n = 5382 \rightarrow \text{even} = 8+2=10; \text{odd} = 5+3=8$.

Exercise 14: Power of a number

Compute a^n (a raised to the power n) without using any library function.

Example: $2^5 = 32$.

Exercise 15: Factorial

Write an algorithm to compute the factorial of n : $n! = n \times (n-1) \times \dots \times 2 \times 1$.

Example: $5! = 120$.

Exercise 16: Divisors of a number

Write an algorithm to find and display all the divisors of a number n .

Example: $n = 28 \rightarrow 1, 2, 4, 7, 14, 28$.

Exercise 17: Sum of the first N odd numbers

Write an algorithm to compute the sum of the first n odd numbers.

Example: $n = 4 \rightarrow 1+3+5+7 = 16$.

Exercise 18: Check if a number is prime

A prime number has no divisors other than 1 and itself. Write an algorithm to test whether a number n is prime or not.

Exercise 19: Prime numbers up to N

Display all prime numbers less than or equal to a number n .

Example: $n = 20 \rightarrow 2, 3, 5, 7, 11, 13, 17, 19$.

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Exercise 20: Greatest Common Divisor (GCD)

Find the GCD of two integers a and b using the Euclidean algorithm.

Example: $\text{GCD}(48, 18) = 6$.

Exercise 21: Co-prime numbers

Two numbers a and b are co-prime if their $\text{GCD} = 1$. Write an algorithm to check if a and b are co-prime.

Exercise 22: Perfect number

A perfect number is equal to the sum of its divisors excluding itself. Write an algorithm to find all the perfect numbers between 1 and n .

Example: $n = 6 \rightarrow 1 + 2 + 3 = 6$.

Exercise 23: Armstrong number

An Armstrong number is equal to the sum of its digits, each raised to the power of the number of digits. Write an algorithm to check whether a number n is Armstrong or not.

Example: $n = 153 \rightarrow 1^3 + 5^3 + 3^3 = 153$.

Exercise 24: Strong number

A strong number n is equal to the sum of the factorials of its digits. Write an algorithm to check whether a number n is strong or not.

Example: $n = 145 \rightarrow 1! + 4! + 5! = 145$.

Exercise 25: Palindrome number

Write an algorithm to check whether an integer number n is palindrome; i.e. reads the same backward and forward.

Example: $n = 121 \rightarrow$ palindrome
 $n = 123 \rightarrow$ not.

Exercise 26: Check if a number is a power of 2

Write an algorithm to test if an integer number n is a power of 2.

Example: $n = 8 \rightarrow$ Yes; $n = 12 \rightarrow$ No.

Exercise 27: Check if a number is Fibonacci

Write an algorithm to verify if a number n is Fibonacci. n is Fibonacci means that $(5n^2 + 4)$ or $(5n^2 - 4)$ is a perfect square.

Example: $n = 21 \rightarrow$ Fibonacci
 $n = 22 \rightarrow$ not.

Exercise 28: Decimal to binary

Write an algorithm to convert a decimal integer number n into its binary equivalent.

Example: $n = 29 \rightarrow 11101_2$.

Exercise 29: Decimal to hexadecimal

Write an algorithm to convert a decimal integer number n into its hexadecimal equivalent.

Example: $n = 255 \rightarrow \text{FF}_{16}$.

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Exercise 30: Swapping

Write an algorithm that carries out a swapping of an integer number n ; i.e. that exchanges the high-weight digit with the low-weight digit of an integer number n .

Example

- | | | |
|-----------------|--------|------------------|
| • Input: | | • Output: |
| ○ $n = 5961$ | -----> | ○ 1965 |
| ○ $n = -18$ | -----> | ○ -81 |
| ○ $n = 723859$ | -----> | ○ 923857 |
| ○ $n = 9$ | -----> | ○ 9 |

Exercise 31:

Write an algorithm that reads an integer number n and counts the frequency of each digit (0...9) in the number using **one (01) counter only**.

Example

- **Input:** $n = 122333$
- **Output:**
 - **Digit 1:** 1 time(s)
 - **Digit 2:** 2 time(s)
 - **Digit 3:** 3 time(s)
- **Input:** $n = 58568$
- **Output:**
 - **Digit 5:** 2 time(s)
 - **Digit 6:** 1 time(s)
 - **Digit 8:** 2 time(s)

Exercise 32: Exponential approximation

Write an algorithm that calculates the approximation of e^n with t terms using **one (01) loop only**, where t is a positive integer number:

$$e^n = 1 + \frac{n^1}{1!} + \frac{n^2}{2!} + \frac{n^3}{3!} + \frac{n^4}{4!} + \dots + \frac{n^t}{t!}$$